

Fisheye Patch-Based Automatic Colorization Method for Anime Line-drawings

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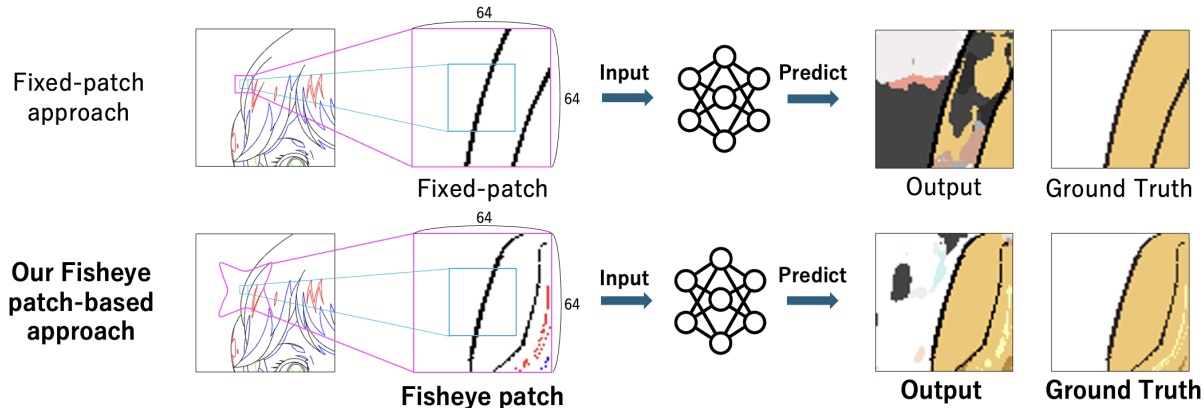
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Figure 1: Workflow of the fisheye patch-based colorization for anime line drawings. Since accurate color prediction from small patches alone is difficult, our method references a broader context beyond the central region to achieve consistent colorization.

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1 INTRODUCTION

Colorizing anime line drawings is still labor-intensive. Patch-based training improves memory efficiency but lacks global context, often confusing similar regions, such as hair and clothing. We propose a *Fisheye Patch Structure* (FPS) that expands visible context without increasing patch size. It preserves high resolution near the center while compressing peripheral areas via distance-aware warping. Under identical training settings, FPS improves accuracy over the standard patch baseline by balancing context and resolution.

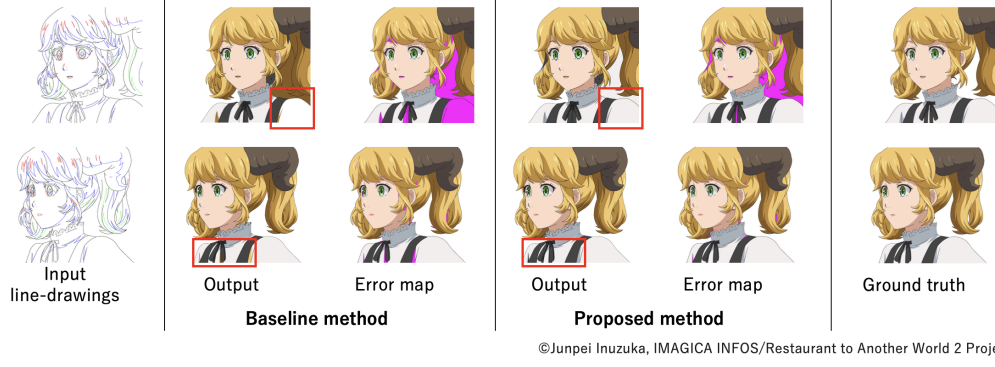


Figure 2: Two examples of colorization results from inference data. For each example, the outputs from the baseline and the proposed method (output) are shown alongside error maps, where miscolored regions compared to the ground truth are highlighted in magenta. The red boxes highlight areas where the proposed method shows notable improvements.

2 FISHEYE PATCH STRUCTURE (FPS)

Figure 1 illustrates the workflow of the proposed fisheye patch structure. Each patch is divided into two parts: a central region, shown as the light-blue square in Figure 1, and a surrounding reference region outside it.

The central region is sampled from the original image at full resolution, while the surrounding reference region is generated by distance-aware resampling of more distant pixels, enabling the model to access a wider context without enlarging the patch size.

Formally, the scaling factor f for each pixel coordinate (x, y) in the patch is defined as follows:

$$f(r_{\text{norm}}) = \begin{cases} 1 & \text{if } (x, y) \in \text{center region} \\ 1 + (s - 1) \cdot \left(1 - e^{-\alpha \cdot r_{\text{norm}}^\gamma}\right) & \text{otherwise} \end{cases}$$

Here, r_{norm} denotes the normalized distance from the boundary of the central region (ranging from 0 to 1), and s , α , and γ are hyperparameters that control the sampling range and density. In our experiments, we empirically set them to $s = 4.0$, $\alpha = 4.0$, and $\gamma = 4.0$ for stable performance.

Using the scaling factor f , each patch coordinate (x, y) is transformed to a sampling location (X, Y) in the original image, with (x_c, y_c) denoting the center of the patch, as follows:

$$(X, Y) = (x_c + f(r_{\text{norm}}) \cdot (x - x_c), y_c + f(r_{\text{norm}}) \cdot (y - y_c)).$$

3 EXPERIMENT AND RESULT

We constructed a dataset using frames from a TV anime, pairing each line drawing with its corresponding color image. Following the few-shot colorization framework of Maejima et al. [Maejima et al. 2024], which colorizes a sequence using only a small number of reference frames, we trained on 3 frames and tested on 6. The model was trained for 3,000 iterations with a batch size of 16, using the network architecture employed in the same study.

We created 64×64 fisheye patches centered on line strokes and trained the model using these patches along with their corresponding ground-truth color patches. During inference, only the central 32×32 region of each patch was used to reconstruct the final image on the original line drawing. For comparison, a baseline model using uniform patches under the same conditions was also trained.

Figure 2 shows example outputs from the test frames. Please refer to the Supplementary Materials for full results. The proposed method effectively reduces common errors observed in the baseline, such as uncolored clothing and miscoloring hair in sparsely lined areas. For quantitative evaluation, we used three metrics: mean Intersection-over-Union (mIoU), Region-wise Accuracy—which checks if each closed region is correctly colorized—and Pixel Accuracy. As shown in Table 1, the proposed method improved all metrics, with mIoU increasing from 89% to 94%.

These results indicate that FPS captured region-specific features more effectively by incorporating broader contextual information than the baseline patch structure, achieving more consistent colorization overall without increasing patch size.

Table 1: Quantitative evaluation results

	Baseline method	Proposed method
mIoU (%)	89.80	94.22
ACC _{region} (%)	79.13	81.67
ACC _{pixel} (%)	93.28	94.55

4 CONCLUSION

We proposed a fisheye patch structure (FPS) that efficiently incorporates broader context without increasing patch size for patch-based colorization of anime line drawings. Experiments demonstrated that FPS outperformed standard rectangular patches, especially reducing miscoloring in regions with sparse line information. This improvement stems from the fisheye transformation, which preserves high-resolution detail at the center while compressively including peripheral context for better region recognition.

In future work, we plan to test FPS on more diverse datasets and evaluate its practical applicability in interactive or production workflows.

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